

Assignment 1: Blockchain-Based Healthcare Data Sharing

The objective of this assignment is to design, develop and evaluate a blockchain-based decentralized application (DApp) that solves a real-world problem. Students are required to implement the system, deploy smart contracts, evaluate blockchain performance and present their work professionally.

Students may use Agentic Coding Tools (e.g., OpenAI Codex, Claude Code, Cursor, Windsurf) to assist with coding, debugging, documentation and testing. However, students must fully understand their implementation and be able to explain every component during the presentation.

Problem Statement:

Healthcare organizations generate large volumes of sensitive patient information. Traditional healthcare information systems are often isolated, making secure data sharing difficult. Centralized storage increases the risk of data breaches, unauthorized access and single points of failure. Patients also have limited control over who can access their medical records.

Blockchain technology combined with decentralized storage can provide secure, transparent, and auditable medical data sharing while allowing patients to control access to their healthcare records.

Students are required to:

1. Develop a blockchain-based healthcare record management system.
2. Implement secure patient-controlled access.
3. Store encrypted medical files.
4. Record medical record ownership and permissions on blockchain.
5. Maintain complete audit logs.
6. Evaluate system performance, scalability, transaction cost and security etc.

Minimum Functional Requirements	Performance Evaluation
● Patient registration ● Doctor registration ● Hospital registration ● Medical record upload ● Record sharing ● Permission granting ● Permission revocation ● Medical history ● Audit log ● Administrator dashboard	Students must calculate: ● Smart contract deployment cost ● Transaction cost ● Gas consumption ● Upload latency ● Download latency ● Permission grant latency ● Permission revoke latency ● Encryption, Decryption time ● Throughput and CPU usage ● Memory usage ● Network bandwidth, Scalability analysis

Important: Students must understand all generated code. During the presentation, students should be able to explain the system architecture, smart contracts, implementation details, and performance evaluation. Failure to explain the project may result in mark deductions.

Final Submission Requirements:

Student must submit:

1. Assignment Report (Hard Copy): The report must not exceed 15 pages.
2. A 10-minute demonstration video, including:
 - Problem statement
 - System architecture
 - Technology stack
 - System demonstration
 - Smart contract overview
 - Performance evaluation (gas cost, latency, throughput, etc.)
 - Challenges and future improvements

Assignment 2: Blockchain-Based Academic Certificate Verification System

The objective of this assignment is to design, develop and evaluate a blockchain-based decentralized application (DApp) that solves a real-world problem. Students are required to implement the system, deploy smart contracts, evaluate blockchain performance and present their work professionally.

Students may use Agentic Coding Tools (e.g., OpenAI Codex, Claude Code, Cursor, Windsurf) to assist with coding, debugging, documentation and testing. However, students must fully understand their implementation and be able to explain every component during the presentation.

Problem Statement:

Academic certificate forgery is a growing problem worldwide. Educational institutions and employers often spend considerable time and effort verifying certificates issued by universities. Traditional centralized databases are vulnerable to unauthorized modifications, data loss, and security breaches. As a result, fake certificates can be difficult to detect.

Blockchain technology offers an immutable and transparent ledger where certificate information can be securely stored. By recording certificate hashes on the blockchain, universities can issue tamper-proof digital certificates that employers and other institutions can verify instantly without relying on a centralized authority.

Students are required to:

1. Develop a blockchain-based certificate verification system.
2. Design and deploy smart contracts for certificate management.
3. Allow universities to issue digital certificates.
4. Allow employers to verify certificate authenticity.
5. Generate QR codes for certificate verification.
6. Prevent certificate tampering.
7. Evaluate blockchain transaction performance, cost etc.

Minimum Functional Requirements	Performance Evaluation
• University registration • Student registration • Certificate issuance • Certificate verification • QR code generation • Certificate revocation • Administrator dashboard	Students must calculate: • Smart contract deployment cost • Gas consumption • Transaction cost (Gas Fee) • Certificate issuance latency • Verification latency • Block confirmation time • Transaction throughput (TPS) • Success rate • Scalability using multiple certificate volumes

Important: Students must understand all generated code. During the presentation, students should be able to explain the system architecture, smart contracts, implementation details, and performance evaluation. Failure to explain the project may result in mark deductions.

Final Submission Requirements:

Student must submit:

3. Assignment Report (Hard Copy): The report must not exceed 15 pages.
4. A 10-minute demonstration video, including:
 - Problem statement
 - System architecture
 - Technology stack
 - System demonstration
 - Smart contract overview
 - Performance evaluation (gas cost, latency, throughput, etc.)
 - Challenges and future improvements

Assignment 3: Blockchain-Based Secure Banking Transaction System

The objective of this assignment is to design, develop and evaluate a blockchain-based decentralized application (DApp) that solves a real-world problem. Students are required to implement the system, deploy smart contracts, evaluate blockchain performance and present their work professionally.

Students may use Agentic Coding Tools (e.g., OpenAI Codex, Claude Code, Cursor, Windsurf) to assist with coding, debugging, documentation and testing. However, students must fully understand their implementation and be able to explain every component during the presentation.

Problem Statement:

Traditional banking systems depend on centralized databases to record financial transactions. These systems may face challenges such as data tampering, delayed transaction verification, limited transparency and dependence on a central authority. Customers also have limited visibility into transaction history and verification processes.

A blockchain-based banking transaction system can provide transparent, immutable and traceable financial records. Every transaction is securely recorded on the blockchain, ensuring data integrity and improving trust between customers and financial institutions.

Students are required to:

1. Develop a blockchain-based banking transaction system.
2. Implement secure account management.
3. Record deposits, withdrawals and transfers using smart contracts.
4. Maintain immutable transaction history.
5. Generate transaction receipts.
6. Evaluate transaction cost, latency, scalability and performance etc.

Minimum Functional Requirements	Performance Evaluation
• Customer registration • Account creation • Deposit • Withdrawal • Money transfer • Transaction history • Account balance inquiry • Administrator dashboard	Students must calculate: • Smart contract deployment cost • Average transaction cost • Gas consumption • Deposit latency • Withdrawal latency • Transfer latency • Block confirmation time • Throughput (Transactions/Second) • CPU, Memory usage • Scalability under different transaction loads

Important: Students must understand all generated code. During the presentation, students should be able to explain the system architecture, smart contracts, implementation details and performance evaluation. Failure to explain the project may result in mark deductions.

Final Submission Requirements:

Student must submit:

1. Assignment Report (Hard Copy): The report must not exceed 15 pages.
2. A 10-minute demonstration video, including:
 - Problem statement
 - System architecture
 - Technology stack
 - System demonstration
 - Smart contract overview
 - Performance evaluation (gas cost, latency, throughput, etc.)
 - Challenges and future improvements